

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé I, §1 — The functors Γ_Z and $\underline{\Gamma}_Z$

English/Germanic language-management lane

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Exposé I

Global and Local Cohomological Invariants Relative to a Closed Subspace

1. The functors Γ_Z and $\underline{\Gamma}_Z$

Let X be a topological space, and let $\underline{\mathcal{C}}_X$ be the category of sheaves of abelian groups on X . Let Φ be a family of supports in the sense of Cartan. The functor Γ_Φ on $\underline{\mathcal{C}}_X$ is defined by

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$$\Gamma_\Phi(F) = \{f \in \Gamma(F) \mid \text{supp}(f) \in \Phi\}. \quad (1)$$

If Z is a closed subset of X , by an abuse of notation we denote by Γ_Z the functor Γ_Φ , where Φ is the family of closed subsets of X contained in Z .¹ Thus

$$\Gamma_Z(F) = \{f \in \Gamma(F) \mid \text{supp}(f) \subset Z\}. \quad (2)$$

We wish to extend this definition to the case in which Z is a *locally closed* subset of X , hence closed in a suitable open subset V of X . In that case we set

$$\Gamma_Z(F) = \Gamma_Z(F|_V). \quad (3)$$

We must verify that $\Gamma_Z(F)$ “does not depend” on the chosen open set. It is enough to show that, if V' is open and $V \supset V' \supset Z$, then the map $\rho_{V'}^V : F(V) \rightarrow F(V')$ maps $\Gamma_Z(F|_V)$ isomorphically onto $\Gamma_Z(F|_{V'})$. But

$$\Gamma_Z(F|_V) = \ker \rho_{V-Z}^V. \quad (4)$$

¹At this single occurrence the French printed and TeX sources underline Γ_Z . The defining equation (2) and the systematic distinction used thereafter identify the intended functor here as the global, non-underlined Γ_Z . The source anomaly and this explicit resolution are recorded in the tranche comparison ledger.

Consequently, if $f \in \Gamma_Z(F|_V)$ and $\rho_{V'}^V(f) = \rho_{V-Z}^V(f) = 0$, then $f = 0$, since $(V', V - Z)$ is a covering of V . Likewise, if $f' \in \Gamma_Z(F|_{V'})$, then $f' \in F(V')$ and $0 \in F(V - Z)$ define an element $f \in F(V)$ such that $\rho_{V'}^V(f) = f'$ and $f \in \Gamma_Z(F|_V)$. Thus $\rho_{V'}^V$ induces an isomorphism

$$\Gamma_Z(F|_V) \longrightarrow \Gamma_Z(F|_{V'}).$$

Note that every open subset W of Z is induced by an open subset U of X in which W is closed. It follows that $W \mapsto \Gamma_W(F)$ defines a presheaf on Z ; one verifies that it is a sheaf, denoted by $i^!(F)$, where $i : Z \rightarrow X$ is the canonical immersion. We obtain

$$\Gamma_Z(F) = \Gamma(i^!(F)). \quad (5)$$

The sheaf $i^!(F)$ is a subsheaf of $i^*(F)$. Indeed, the canonical homomorphism

$$\Gamma(F|_U) = \Gamma(U, F) \longrightarrow \Gamma(U \cap Z, i^*(F))$$

is injective on $\Gamma_{U \cap Z}(F|_U) \subset \Gamma(F|_U)$. In summary, we have the following result.

Proposition 1.1. *There exists a unique subsheaf $i^!(F)$ of $i^*(F)$ such that, for every open subset U of X for which $U \cap Z$ is closed in U , the map*

$$\Gamma(F|_U) = \Gamma(U, F) \longrightarrow \Gamma(U \cap Z, i^*(F))$$

induces an isomorphism

$$\Gamma_{U \cap Z}(F|_U) \longrightarrow \Gamma(U \cap Z, i^!(F)).$$

If Z is open, then simply

$$i^!(F) = i^*(F) = F|_Z, \quad \Gamma_Z(F) = \Gamma(Z, F). \quad (6)$$

Suppose again that Z is arbitrary. As U ranges through the open subsets of X , the assignment

$$U \longmapsto \Gamma_{U \cap Z}(F|_U) = \Gamma(U \cap Z, i^!(F))$$

is a sheaf on X , which we denote by $\underline{\Gamma}_Z(F)$. More precisely, the preceding formula, which expresses the fact that $i^!$ commutes with restriction to open subsets, gives an isomorphism

$$\underline{\Gamma}_Z(F) = i_*(i^!(F)). \quad (7)$$

By definition, for every open subset U of X ,

$$\Gamma(U, \underline{\Gamma}_Z(F)) = \Gamma_{U \cap Z}(F|_U). \quad (8)$$

Here there is a characteristic difference between the case in which Z is closed and the case in which Z is open. In the first case, formula (8) shows that $\underline{\Gamma}_Z(F)$ may be regarded as a subsheaf of F ; hence there is a *canonical immersion*

$$\underline{\Gamma}_Z(F) \hookrightarrow F. \quad (8')$$

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When Z is open, on the other hand, equation (6) shows that the right-hand side of equation (8) is $\Gamma(U \cap Z, F)$ and therefore receives a homomorphism from $\Gamma(U, F)$. Thus there is a *canonical homomorphism* in the opposite direction:²

$$F \longrightarrow \Gamma_Z(F). \quad (8'')$$

This is none other than the canonical homomorphism

$$F \longrightarrow i_* i^*(F),$$

in view of the isomorphism

$$\Gamma_Z(F) \simeq i_* i^*(F) \quad (6 \text{ bis})$$

deduced from equations (6) and (7).

For variable F , the objects $\Gamma_Z(F)$, $\Gamma_Z(F)$, and $i^!(F)$ may of course be regarded as functors in F , with values, respectively, in the category of abelian groups, the category of sheaves of abelian groups on X , and the category of sheaves of abelian groups on Z . It is sometimes convenient to interpret the functor

$$i^! : \underline{\mathbb{C}}_X \longrightarrow \underline{\mathbb{C}}_Z$$

as the right adjoint of the familiar functor

$$i_! : \underline{\mathbb{C}}_Z \longrightarrow \underline{\mathbb{C}}_X$$

defined by the following proposition.

Proposition 1.2. *Let G be a sheaf of abelian groups on Z . There exists a unique subsheaf of $i_*(G)$, denoted $i_!(G)$, such that for every open subset U of X the (identity) isomorphism*

$$\Gamma(U \cap Z, G) = \Gamma(U, i_*(G))$$

induces an isomorphism

$$\Gamma_{\Phi_{U \cap Z, U}}(U \cap Z, G) = \Gamma(U, i_!(G)),$$

where $\Phi_{U \cap Z, U}$ denotes the family of subsets of $U \cap Z$ that are closed in U .

The verification reduces to observing that the left-hand side is a sheaf as U varies: namely, for a section of $i_*(G)$ on U , viewed as a section of G on $U \cap Z$, the property of having support closed in U is *local on U* . The sheaf $i_!(G)$ just defined is also known as the *sheaf obtained from G by extension by zero outside Z* ; see Godement [Godement]. In particular, if Z is closed, then

$$i_!(G) = i_*(G). \quad (9)$$

In general, however, the canonical injection $i_!(G) \rightarrow i_*(G)$ is not an isomorphism, as is already well known when Z is open. Evidently $i_!(G)$ depends functorially on G (and is in fact an exact functor of G). We therefore have:

²The original reference was (8).

Proposition 1.3. *There is an isomorphism of bifunctors in G and F , where G is a sheaf of abelian groups on Z and F is a sheaf of abelian groups on X :*

$$\mathrm{Hom}(i_!(G), F) = \mathrm{Hom}(G, i^!(F)). \quad (10)$$

To define such an isomorphism is equivalent to defining functorial homomorphisms

$$i_! i^!(F) \longrightarrow F, \quad G \longrightarrow i^! i_!(G),$$

satisfying the familiar compatibility conditions (see, for example, Shih's exposé in the Cartan seminar on cohomological operations).

Recalling that $i_!$ is exact, and hence takes monomorphisms to monomorphisms, we conclude:

Corollary 1.4. *If F is injective, then $i^!(F)$ is injective; consequently $\Gamma_Z(F) = i_* i^!(F)$ is also injective.*

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Replacing X by a variable open subset U of X , Proposition 1.3 also gives:

Corollary 1.5. *There is an isomorphism, functorial in F and G ,*

$$\underline{\mathrm{Hom}}(i_!(G), F) = i_*(\underline{\mathrm{Hom}}(G, i^!(F))). \quad (11)$$

Taking G to be the constant sheaf on Z defined by $\mathbf{Z}$, denoted $\mathbf{Z}_Z$, Proposition 1.3 and Corollary 1.5 specialize as follows.

Corollary 1.6. *There are isomorphisms, functorial in F ,*

$$\begin{aligned} \Gamma_Z(F) &= \mathrm{Hom}(\mathbf{Z}_{Z,X}, F), \\ \Gamma_Z(F) &= \underline{\mathrm{Hom}}(\mathbf{Z}_{Z,X}, F), \end{aligned} \quad (12)$$

where $\mathbf{Z}_{Z,X} = i_!(\mathbf{Z}_Z)$ is the sheaf of abelian groups on X obtained from the constant sheaf $\mathbf{Z}_Z$ on Z by extension by zero outside Z .

Remark 1.7. Suppose that X is a ringed space, equip Z with the sheaf of rings $\mathcal{O}_Z = i^{-1}(\mathcal{O}_X)$, and now let $\underline{\mathbf{C}}_X$ and $\underline{\mathbf{C}}_Z$ denote the categories of modules on X and Z , respectively. The preceding considerations extend word for word, with F a module on X and G a module on Z , and with statements 1.3–1.6 interpreted accordingly.

To conclude these generalities, let us examine what happens when the locally closed subset Z is changed. Let $Z' \subset Z$ be another locally closed subset, and let

$$j : Z' \longrightarrow Z, \quad i' : Z' \longrightarrow X, \quad i' = i j$$

be the canonical inclusions. There are functorial isomorphisms

$$(ij)^! = j^! i^!, \quad (ij)_! = i_! j_!. \quad (13)$$

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The first isomorphism in (13) defines a functorial isomorphism

$$\Gamma_{Z'}(F) = \Gamma(Z', (ij)^!(F)) \simeq \Gamma(Z', j^!(i^!(F))) = \Gamma_{Z'}(i^!(F)). \quad (14)$$

Suppose now that Z' is *closed* in Z , and let

$$Z'' = Z - Z'$$

be its complement in Z . This is open in Z and hence locally closed in X . Applying the canonical injection (8') to $i^!(F)$ on Z with the subset Z' , and using (14), gives an injective functorial canonical homomorphism

$$\Gamma_{Z'}(F) \longrightarrow \Gamma_Z(F). \quad (15)$$

If Z' is replaced by Z'' in (14) and equation (8'') is used, one obtains a functorial canonical homomorphism

$$\Gamma_Z(F) \longrightarrow \Gamma_{Z''}(F). \quad (15')$$

Proposition 1.8. *Under the preceding hypotheses, the sequence of functorial homomorphisms*

$$0 \longrightarrow \Gamma_{Z'}(F) \longrightarrow \Gamma_Z(F) \longrightarrow \Gamma_{Z''}(F) \quad (16)$$

is exact. If F is flasque, the sequence remains exact after a zero is added on the right.

Proof. Replace X by an open subset V in which Z is closed. This reduces the assertion to the case in which Z , and hence Z' , is closed. Then Z'' is closed in the open subset $X - Z'$, and there is a canonical injection

$$\Gamma_{Z''}(F) \longrightarrow \Gamma(X - Z', F).$$

The exactness of (16) then says simply that the sections of F with support in Z' are precisely those whose restriction to $X - Z'$ is zero.

If F is flasque, every element of $\Gamma_{Z''}(F)$, regarded as a section of F on $X - Z'$, extends to a section of F on X . The latter section plainly has support in Z , proving that the last homomorphism in (16) is then surjective. $\square$

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Corollary 1.9. *There is a functorial exact sequence*

$$0 \longrightarrow \underline{\Gamma}_{Z'}(F) \longrightarrow \underline{\Gamma}_Z(F) \longrightarrow \underline{\Gamma}_{Z''}(F), \quad (16 \text{ bis})$$

and, if F is flasque, the sequence remains exact after a zero is added on the right.

Proposition 1.8 may be interpreted, by means of Corollary 1.6, as a statement about the functors Hom and $\underline{\text{Hom}}$, as follows. First note that if G is a sheaf of abelian groups on Z , inducing the sheaves $j^*(G)$ and $k^*(G)$ on Z' and Z'' , respectively, where $j : Z' \rightarrow Z$ and $k : Z'' \rightarrow Z$ are the canonical injections, then there is a canonical exact sequence of sheaves on X :

$$0 \longrightarrow k^*(G)_X \longrightarrow G_X \longrightarrow j^*(G)_X \longrightarrow 0. \quad (17)$$

Here, to simplify notation, the subscript X denotes the sheaf on X obtained by extension by zero outside the space on which the sheaf in question is defined. Sequence (17) generalizes a familiar exact sequence for $Z = X$ (see Godement [Godement]); it follows from that sequence by writing it on Z and applying the functor $i_!$. Taking $G = \mathbf{Z}_Z$, we obtain in particular:

Proposition 1.10. *Under the preceding hypotheses, there is an exact sequence of sheaves of abelian groups on X :*

$$0 \longrightarrow \mathbf{Z}_{Z'',X} \longrightarrow \mathbf{Z}_{Z,X} \longrightarrow \mathbf{Z}_{Z',X} \longrightarrow 0. \quad (18)$$

The two exact sequences in Proposition 1.8 and Corollary 1.9 are exactly the sequences obtained from (18) by applying, respectively, the functors $\mathrm{Hom}(-, F)$ and $\underline{\mathrm{Hom}}(-, F)$.

This also gives an immediate proof that sequences (16) and (16 bis) remain exact after a zero is added on the right, provided that F is *injective*.

References

[Godement] R. Godement, *Topologie algébrique et théorie des faisceaux*, Act. Scient. et Ind., vol. 1252, Hermann, Paris.